

The Virtual-Augmented Reality Simulator: Evaluating OST-HMD AR calibration algorithms in VR

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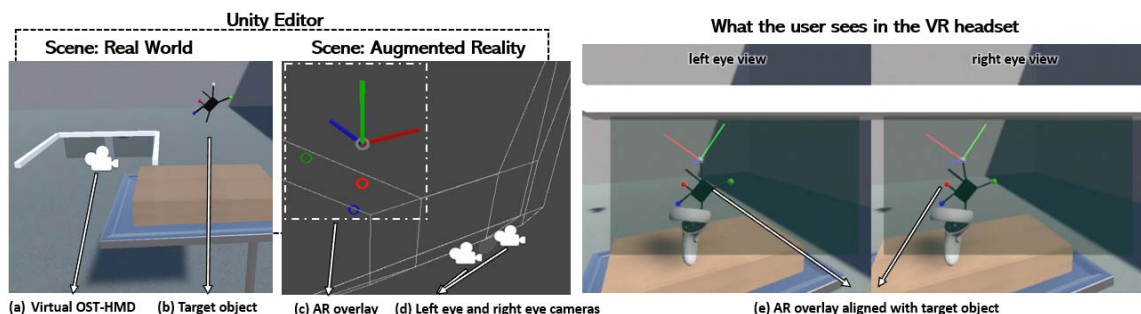
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Figure 1: How it works: to simulate AR in VR, our simulator uses two scenes. The “Real World” is a VR scene much like any VR application. In it, the simulator adds a virtual OST-HMD with two displays. The Augmented Reality scene is a separate scene that only contains the AR part of the simulation. For every frame rendered, the OST-HMD *pose* is sent to the Augmented Reality scene. Based on the user’s inter-pupillary distance, two cameras - one per eye - follow the pose of the virtual OST-HMD to render one AR view per display. The poses of other objects can also be sent and used in the Augmented Reality scene. **Left:** Simulated real world scene – (a) Virtual OST-HMD with two displays. (b) Target object to be augmented in AR. **Middle:** Augmented Reality scene – This scene is designed completely isolated from the real-world environment and can be re-used with different environments. (c) Example of an AR visualization to be aligned with the target object in the real world scene. (d) left and eye cameras responsible for rendering the AR overlay scene. **Right:** Virtual Reality user view showing a preview of both eyes – (e) rendering on both displays.

ABSTRACT

When developing AR applications for high-precision domains such as surgery, we face a common problem: how can the system guarantee that the end-user will see a virtual object aligned with its real-world counterpart? Alignment, or registration, is a crucial feature of AR displays, but achieving accurate alignment between real and virtual objects is not trivial. With hundreds of calibration approaches available, we need better tools to understand how and when calibration algorithms fail as well as understand what can be done to improve alignment. This poster introduces a novel AR simulator in VR that facilitates experimentation with different calibration algorithms.

Keywords: Augmented Reality, Head-Mounted Displays, Alignment, Registration, Simulator, Virtual Reality

Index Terms: Augmented Reality—Head-Mounted Displays—Alignment—Simulator;

1 INTRODUCTION

Alignment, or registration, is as a key feature of AR displays [2], but achieving accurate alignment between real and virtual objects is not trivial [5], and it usually requires one or more *calibration* steps. Over the years, our research community has thrived in creating and

improving calibration methods that improve alignment. A recent survey by Grubert et al., for example, lists 270 different calibration algorithms proposed until 2017 [4]. With these many options available, it can be difficult to select a method most applicable to a specific headset and application.

Implementing and experimenting with various calibration methods requires extensive instrumentation of the environment and the user, but even when these are available, we lack tools that can help us understand trade-offs between different calibration algorithms. Moreover, the alignment error is a combination of errors arising from different sources, so it is hard to pin-point what calibration or aspect should be improved next: optical distortions from the display technology? Poor user-display setup (e.g., wrong IPD, poorly worn head mounted display)? Drift or occlusion introduced by tracking system? Numerical instability from mathematical models?

With this work, we argue that our community needs better tools to support us in using and evaluating calibration algorithms. To address the current complexity of the alignment process, and the difficulties that this creates for development, we introduce an AR simulator in Virtual Reality. Our simulator provides an interface to implement and test calibration setups without the need to deploy to the real AR HMD nor the need to instrument the environment with expensive tracking technology. Moreover, unlike real life, we can always obtain ground-truth measurements in the simulator.

This poster introduces the The Virtual-Augmented Reality Simulator: an interactive Optical See-Through (OST) head-mounted display (HMD) simulator that allows developers and researchers to control the different parameters that influence alignment, compare calibration results with ground-truth values, and - when necessary - run user experiments in Virtual Reality.

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2 RELATED WORK

In the past, researchers have successfully used simulated AR in Virtual Reality for calibration purposes [3, 6]. For example, Martin et al. [6] studied the task of aligning a 3D object over another 3D object in Virtual Reality. Today, however, there aren't tools that can simulate a generic OST-HMD in Virtual Reality. Our work fills this gap with an open-source, customizable OST-HMD simulator.

3 THE VIRTUAL-AUGMENTED REALITY SIMULATOR

Our VR OST-HMD simulator creates a virtual augmented reality headset in a virtual environment. Similar to real life, the head-mounted display in this virtual environment is a pair of goggles that users can wear to experience AR. The virtual OST-HMD renders a different picture per eye, so that the user can see and interact with them in three dimensions (Fig. 1).

One of the main benefits of simulating both real and augmented environments is that we have control over both environments as well as how and when information traverses from one to the other. For example, the simulator guarantees that all tracking data from the simulated real world is used by the virtual OST-HMD at the same point in time, within the same frame.

3.1 Simulator's Features

Alignment is influenced by many factors. To simulate a variety of alignment workflows, our virtual AR simulator provides researchers and developers with the following features:

(1) Virtual Trackers - A core building block of our system is a virtual tracking system that broadcasts the *pose* of objects of interest in the simulated real scene to the augmented reality scene. This includes the OST-HMD position and the location of its displays. Trackers can either share a global coordinate system center or belong to a shared coordinate system centered at a specific tracker. *Virtual trackers* can be customized in many ways: tracker drift (e.g., constant, per acceleration, per time-step, based on a curve), jitter (e.g., Gaussian), and latency (e.g., constant, distance-based, Gaussian). Moreover, they can also be customized through user-defined callbacks. Finally, *virtual trackers* can be referenced by the AR scene by name.

(2) Locatable Cameras - Locatable cameras are cameras that, for each frame captured, attach a transformation matrix that maps points from the camera coordinate system to the head-mounted display coordinate system. They are commonly available in modern HMDs such as Microsoft HoloLens. Researchers have used this camera, for example, to track and augment target objects visible to the user. We provide a locatable camera with our virtual OST-HMD that developers can use to test and implement computer-vision tracking methods. They can also control the camera's resolution, field-of-view, and its location on the OST-HMD.

(3) Linear Equation Solver - Generally, the goal of calibration algorithms is to find a perspective, affine or orthogonal transformation between two coordinate systems. This transformation is organized as a system of linear equations and solved through point correspondences between the coordinates systems [4]. Our simulator includes a basic generic solver that can produce perspective, affine or orthogonal transformations given a list of point correspondences.

(4) Customizable OST-HMD - All aspects of the OST-HMD are customizable: display position, rotation, size, resolution, and its projection matrix. This flexibility of customization allows us to create simulations for modern OST-HMDs (e.g., HoloLens and Magic Leap) as well as experiment with new displays. Through post-processing shader, we can also simulate color bleeding, optical distortion, etc.

(5) Template scenarios - To help other researchers get started with our simulator, we provide three templates scenarios: **(a) 3D-3D scenario** in which a user maps points from the OST-HMD 3D coordinate system to another 3D coordinate system (e.g., a tracked

object [8]); **(b) locatable camera scenario** a template that shows how to use the virtual camera to track fiducial markers such as ArUco Markers [7]); **(c) 3D-2D calibration**: a variation of the Single Point Active Alignment Method (SPAAM) [9].

3.2 Implementation

Today, a commonly used environment for developing Augmented Reality applications is Unity [1]. Given that Unity supports both VR and AR, we decided to use Unity as a platform for our Virtual AR Simulator. Developing a virtual-augmented reality application for our simulator is no different than developing an AR application for modern OST-HMDs. Moreover, the simulated AR scene is set in a way that allows us to reuse code when deploying to an real OST-HMD (and vice-versa). Because Unity provides limited support for solving systems of linear equations, we rely on a custom C++ OpenCV plugin for our Linear Equation Solver; however, we also implement a Unity script to facilitate its use by developers. The simulator is available as an open source Unity package at <https://github.com/WeibelLab/HMDSimulator> under the MIT License.

4 SUMMARY AND FUTURE WORK

In this poster, we introduced a Virtual Reality Simulator for evaluating augmented reality alignment workflows. Alignment workflows support the task of aligning a virtual object displayed on the HMD with a real-world target. They are typically composed of a complex combination of OST-HMD parameters, user interaction, tracking methods, and calibration methods.

To better understand how measurements obtained in the simulator predict real-life results, we are currently conducting a comparative user study between an alignment workflow measured in the simulator with one deployed to a modern AR OST-HMD. While the initial results show a gap in performance between the simulated display and the real display, the goal is to use the simulator to introduce errors and then measure how different sources of errors (e.g., display optical distortion, accommodation-vergence conflict, tracking drift) impact the overall alignment error (position and orientation).

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